

中国科学院大学
University of Chinese Academy of Sciences

Generative Models & AI4S Applications

Lecture 2 Probabilistic Modeling

Fall 2026

Liu Yunfan (刘云帆)

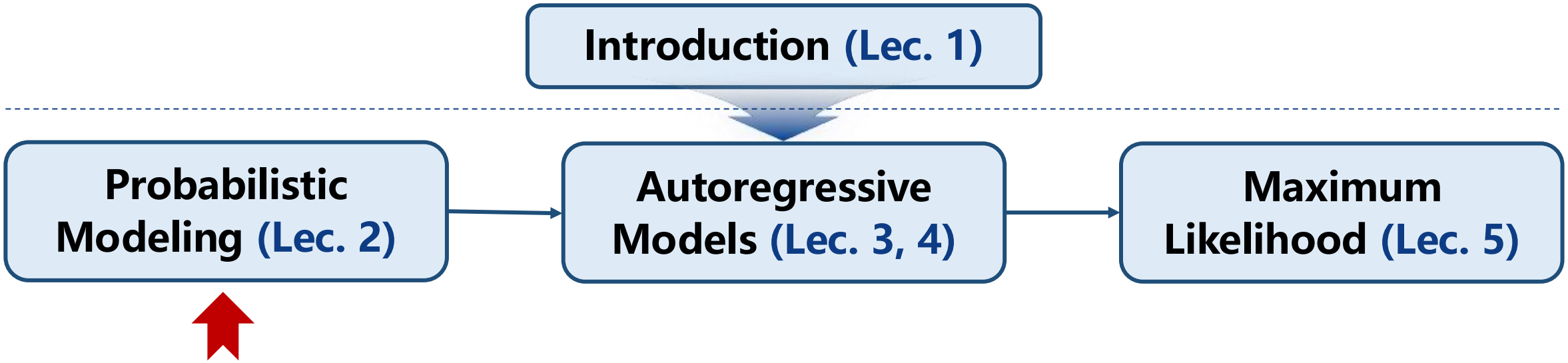
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Announcements

- Make sure to get in the **WeChat group** of this course!
  - ➢ And be sure to change your nickname to **[name]-[number]**

- Tutorial on the usage of Kaggle Platform and GPU acceleration is uploaded online

Roadmap



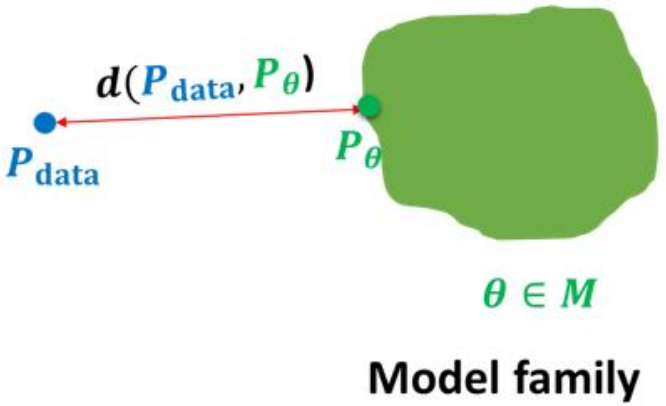
Formulating the probabilistic generative model p_θ

Recap: Learning a Generative Model

Images of dogs



$x_i \sim P_{\text{data}}$
$i = 1, 2, \ldots, n$



Dog Image Generator

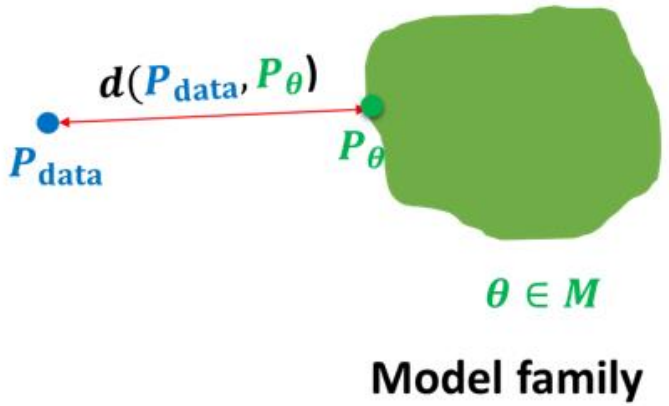
- **Goal:** learn a probability distribution $p_\theta(x)$ over images x
  - **Generation:** Sample $x_{new} \sim p_\theta(x)$, x_{new} should "look like a dog"
  - **Likelihood Estimation:** $p_\theta(x_0)$ should be high is x_0 looks like a dog
  - **Representation:** Discover common features, e.g., ears, tails, etc.
- **Question: How to learn $p_\theta(x)$ as a Statistical ML model?**

Recap: Principles of Statistical Machine Learning

Images of dogs



$x_i \sim P_{\text{data}}$
$i = 1, 2, \ldots, n$



Dog Image Generator

- Three main components of a Statistical ML algorithm:
  - **Model Family (模型):** Define the scope of learnable function (i.e., $P_\theta(x)$)
  - **Learning Objective (准则):** Quantify the target of learning process $\mathcal{L}(x; \theta)$
  - **Optimization (算法):** Algorithm for solving the optimal parameters θ^*

$$\theta^* = \arg\min_\theta \mathbb{E}_{x \sim p_{\text{data}}(x)}[\mathcal{L}(x; \theta)]$$

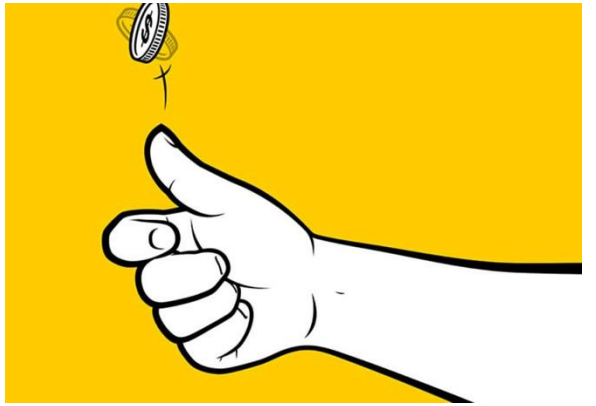


Lecture Outline

- **Probabilistic Models of the Data**
- **Generative vs. Discriminative Models**
- **Deep Generative Models**

Probabilistic Models of the Data

- **Basic (Discrete) Distributions**
  - **Bernoulli distribution: (biased) Coin flipping**
    - ✓ $X \sim \{\text{Heads}, \text{Tails}\}$
    - ✓ $P(X = \text{Heads}) = p, \;\; P(X = \text{Tails}) = 1 - p$
    - ✓ $X \sim \text{Ber}(p)$ ➔ Sample X by flipping a coin



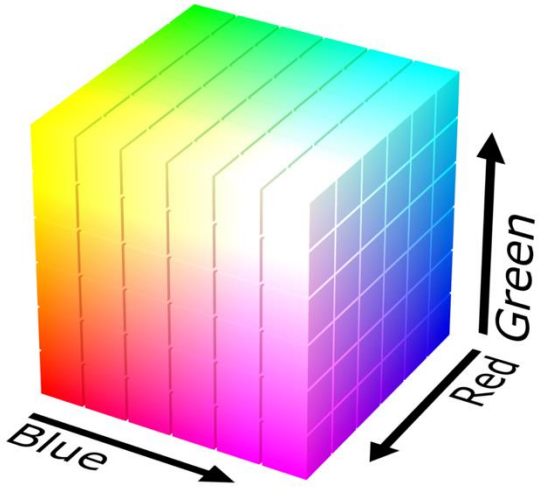
  - **Categorical distribution: (biased) Die rolling**
    - ✓ $Y \sim \{1, 2, \ldots, m\}$
    - ✓ $P(Y = \text{i}) = p_i, \;\; \text{s.t.}, \; \sum p_i = 1$
    - ✓ $Y \sim \text{Cat}(p_1, p_2, \ldots, p_m)$ ➔ Sample Y by rolling a die



  - **How many parameters are need to represent each distribution?**

Probabilistic Models of the Data

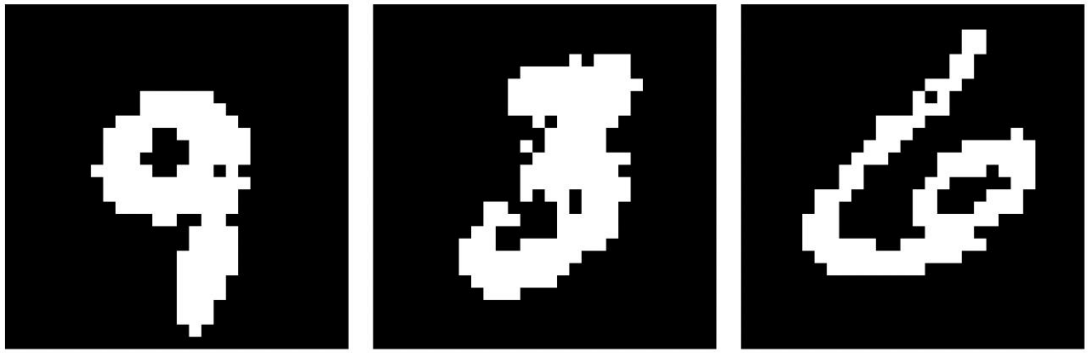
- **A Multi-variate Joint Distribution**
  - **Distribution of a single pixel (R, G, B)**
    - ✓ Red Channel: $\mathrm{Val}(R) = \{0, 1, \dots, 255\}$
    - ✓ Green Channel: $\mathrm{Val}(G) = \{0, 1, \dots, 255\}$
    - ✓ Blue Channel: $\mathrm{Val}(B) = \{0, 1, \dots, 255\}$



  - Sample $(r, g, b) \sim p(R, G, B)$ ➔ A random color for a pixel
  - How many parameters (p_i) are needed to specify the joint distribution $p(R = r, G = g, B = b)$?

Probabilistic Models of the Data

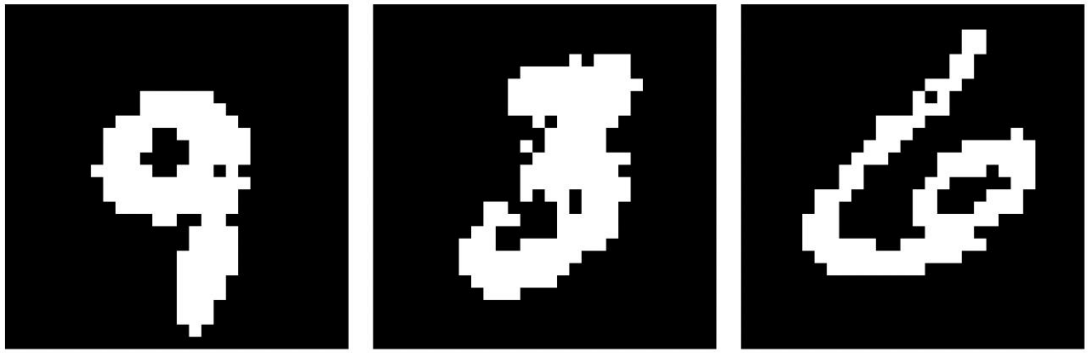
- **Curse of Dimensionality**
  - Consider the distribution of a **binary image** with $n = 28 \times 28$ pixels



  - Each pixel X_i is a binary random variable, $\text{Val}(X_i) = \{0, 1\} = \{\text{Black}, \text{White}\}$
  - How many possible states? $2 \times 2 \times \cdots \times 2 = 2^n$
  - How many parameters to specify the joint distribution $p(x_1, \ldots, x_n)$?

Probabilistic Models of the Data

- **Curse of Dimensionality**
  - Consider the distribution of a **binary image** with $n = 28 \times 28$ pixels

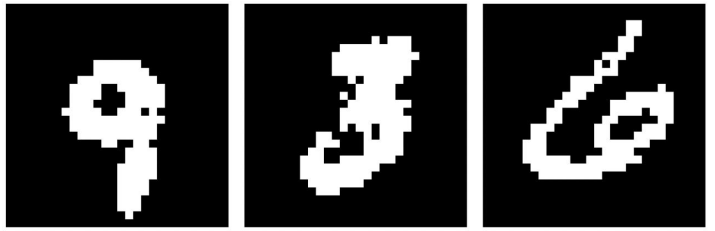


  - Each pixel X_i is a binary random variable, $\mathrm{Val}(X_i) = \{0, 1\} = \{\mathrm{Black}, \mathrm{White}\}$
  - How many possible states? $2 \times 2 \times \cdots \times 2 = 2^n$
  - How many parameters to specify the joint distribution $p(x_1, \ldots, x_n)$?

$2^n - 1$ **Decompose for tractability**

Probabilistic Models of the Data

- **Assumptions on the Joint Distribution**



  - If $X_1, X_2, \ldots, X_n$ are **independent**, then

$$p(x_1, \ldots, x_n) = p(x_1)p(x_2) \ldots p(x_n)$$

  - How many possible states?

$$2 \times 2 \times \cdots \times 2 = 2^n$$

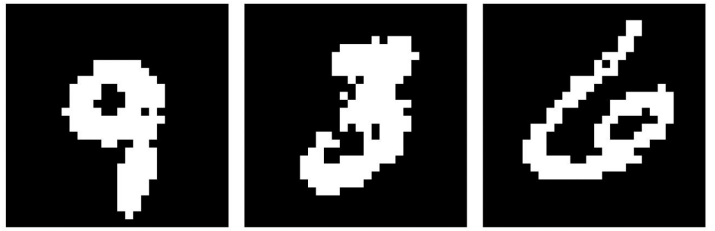
  - How many parameters to specify the joint distribution $p(x_1, \ldots, x_n)$?

$2^n - 1$ ➡ n

Why? At what cost?

Probabilistic Models of the Data

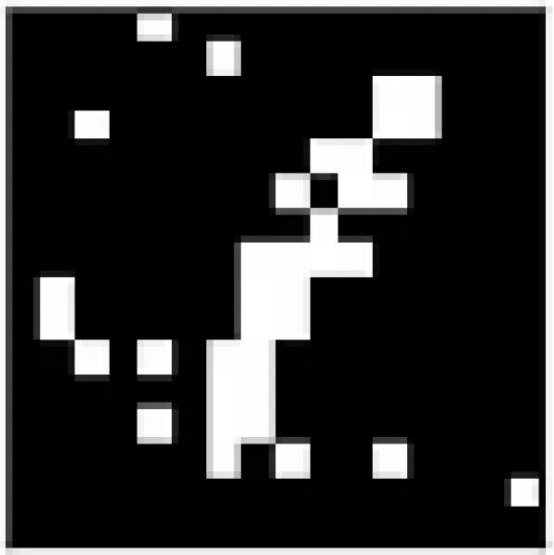
- **Assumptions on the Joint Distribution**

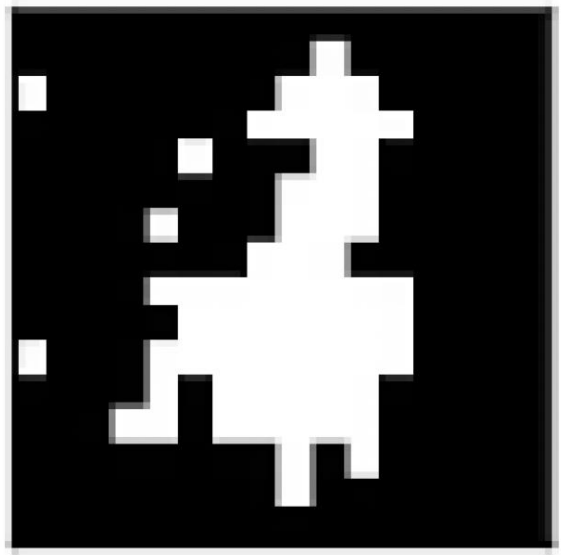


  - If $X_1, X_2, \ldots, X_n$ are **independent**, then

$$p(x_1, \ldots, x_n) = p(x_1)p(x_2) \ldots p(x_n)$$

  - Independence assumption is too strong ➔ **Conditional Independence!**





Sampling results given independent $X_1, X_2, \ldots, X_n$

Review: Probability Theory

- **Conditional Independence**
  - Two events A, B are **conditionally independent** given event C if

$$p(A \cap B \mid C) = p(A \mid C)\, p(B \mid C)$$

  - Random variables X, Y are **conditionally independent** given Z if

$$p(X = x \cap Y = y \mid Z = z) = p(X = x \mid Z = z)\, p(Y = y \mid Z = z)$$

  for all possible values for X, Y and Z

  - **Equivalent notations:**
    - ✓ $p(X, Y \mid Z) = p(X \mid Z)\, p(Y \mid Z)$
    - ✓ $X \perp Y \mid Z$

Review: Probability Theory

- **Chain Rule**
  - Let $S_1, \ldots, S_n$ denote n events, $p(S_i) > 0$

$$p(S_1 \cap S_2 \cap \cdots \cap S_n) = p(S_1)\ p(S_2 \mid S_1)\ \cdots\ p(S_n \mid S_1 \cap \cdots \cap S_{n-1})$$

- **Bayes' Rule**
  - Let $S_1, \ldots, S_n$ denote n events, $p(S_1) > 0$ and $p(S_2) > 0$

$$p(S_1 \mid S_2) = \frac{p(S_1 \cap S_2)}{p(S_2)} = \frac{p(S_2 \mid S_1)\, p(S_1)}{p(S_2)} = p(S_2 \mid S_1)\frac{p(S_1)}{p(S_2)}$$

Probabilistic Models of the Data

- **Conditional Independence**



$$p(x_1, \dots, x_n) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_1, x_2) \cdots p(x_n \mid x_1, \dots, x_{n-1})$$

➢ How many parameters are needed? ($\mathrm{Val}(X_i) = \{0, 1\} = \{\mathrm{Black}, \mathrm{White}\}$)

$$\mathbf{1 + 2 + \cdots + 2^{n-1} = 2^n - 1}$$

Still exponential → No gain from Chain Rule

➢ **Assume** $X_{i+1} \perp X_1, \dots, X_{i-1} \mid X_i$, we have

$$p(x_1, \dots, x_n) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid \cancel{x_1}, x_2) \cdots p(x_n \mid \cancel{x_1, \dots}, x_{n-1})$$

$$= p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2) \cdots p(x_n \mid x_{n-1})$$

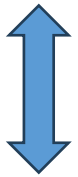
Reduces "expressiveness" !

Probabilistic Models of the Data

- **Conditional Independence**
  - ➢ **Assume** $X_{i+1} \perp X_1, \ldots, X_{i-1} \mid X_i$ …
    - ✓ How many parameters are need to represent $p(x_1, \ldots, x_n)$?

$$p(x_1, \ldots, x_n) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2) \cdots p(x_n \mid x_{n-1})$$

$\boldsymbol{2n-1}$



Conditional Independence

➔ Exponential Reduction

$$p(x_1, \ldots, x_n) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_1, x_2) \cdots p(x_n \mid x_1, \ldots, x_{n-1})$$

$\boldsymbol{2^n-1}$

Reduces “expressiveness”

Probabilistic Models of the Data

- **Conditional Independence**

  - **Assume** $X_{i+1} \perp X_1, \ldots, X_{i-1} \mid X_i$ …

    - ✓ How many parameters are need to represent $p(x_1, \ldots, x_n)$?

$$p(x_1, \ldots, x_n) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2) \cdots p(x_n \mid x_{n-1})$$

$\boldsymbol{2n-1}$

Conditional Independence

➔ Extra parameters

$$p(x_1, \ldots, x_n) = p(x_1)p(x_2) \ldots p(x_n)$$

$\boldsymbol{n}$

Models interdependencies

Probabilistic Models of the Data

- **Bayesian Networks**
  - **Motivation:** Use conditional parameterization (instead of joint)

$$p(x_1, \dots, x_n) = \prod_i p(x_i \mid x_{A_i})$$

where $\boldsymbol{X_{A_i}}$ are **assumed** conditional variables for each X_i ("**Ancestor**")

  - **Note:**
    - ✓ $\prod_i p(x_i \mid x_{A_i})$ has to correspond to a **legal** prob distribution (i.e., no loop)
    - ✓ Independency relationships $\boldsymbol{X_{A_i}}$ reflect **prior knowledge**

Chain Rule ➔ Conditional Independencies ➔ Simplified Factorization

Probabilistic Models of the Data

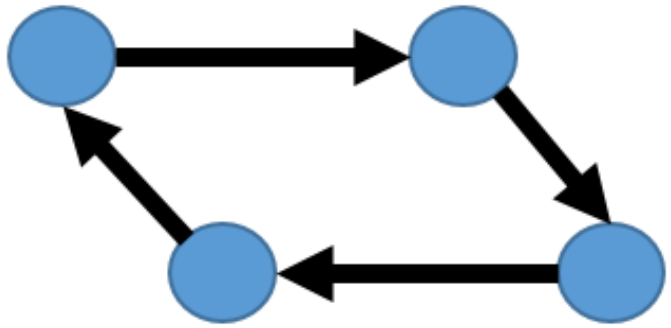
- **Bayesian Networks**
  - **Representation:** A **Directed Acyclic Graph (DAG)** , $G = (V, E)$
    - ✓ One node $i \in V$ for each random variable X_i
    - ✓ **Conditional Probability Distribution (CPD)** per node, $p(x_i \mid \boldsymbol{x}_{Pa(i)})$
      - ➔ X_i's probability conditioned on its parents' values ($\boldsymbol{X}_{Pa(i)}$)
  - **Definition:** a joint distribution

$$p(x_1, \dots x_n) = \prod_{i \in V} p(x_i \mid \boldsymbol{x}_{Pa(i)})$$

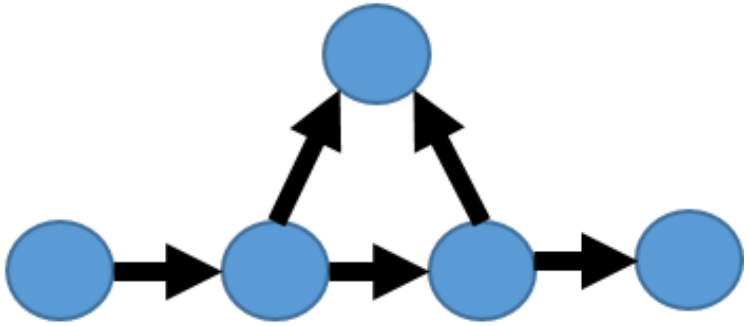
$$\#\text{parameters} = O(|V| \cdot 2^k), \; k = \max_i |Pa(i)| \quad \Leftrightarrow \quad \#\text{parameters} = O(2^{|V|})$$

Probabilistic Models of the Data

- **Bayesian Networks: Directed Acyclic Graph (DAG)**

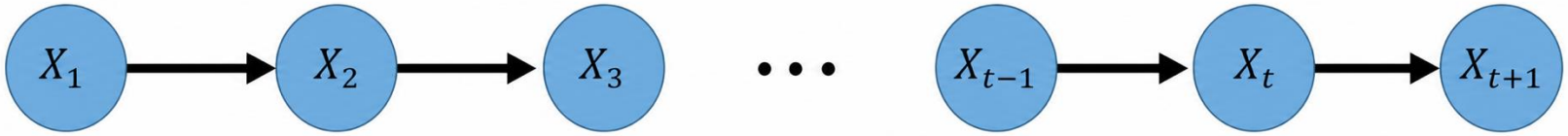


Directed cycle



DAG

$$X_{i+1} \perp X_1, \ldots, X_{i-1} \mid X_i$$



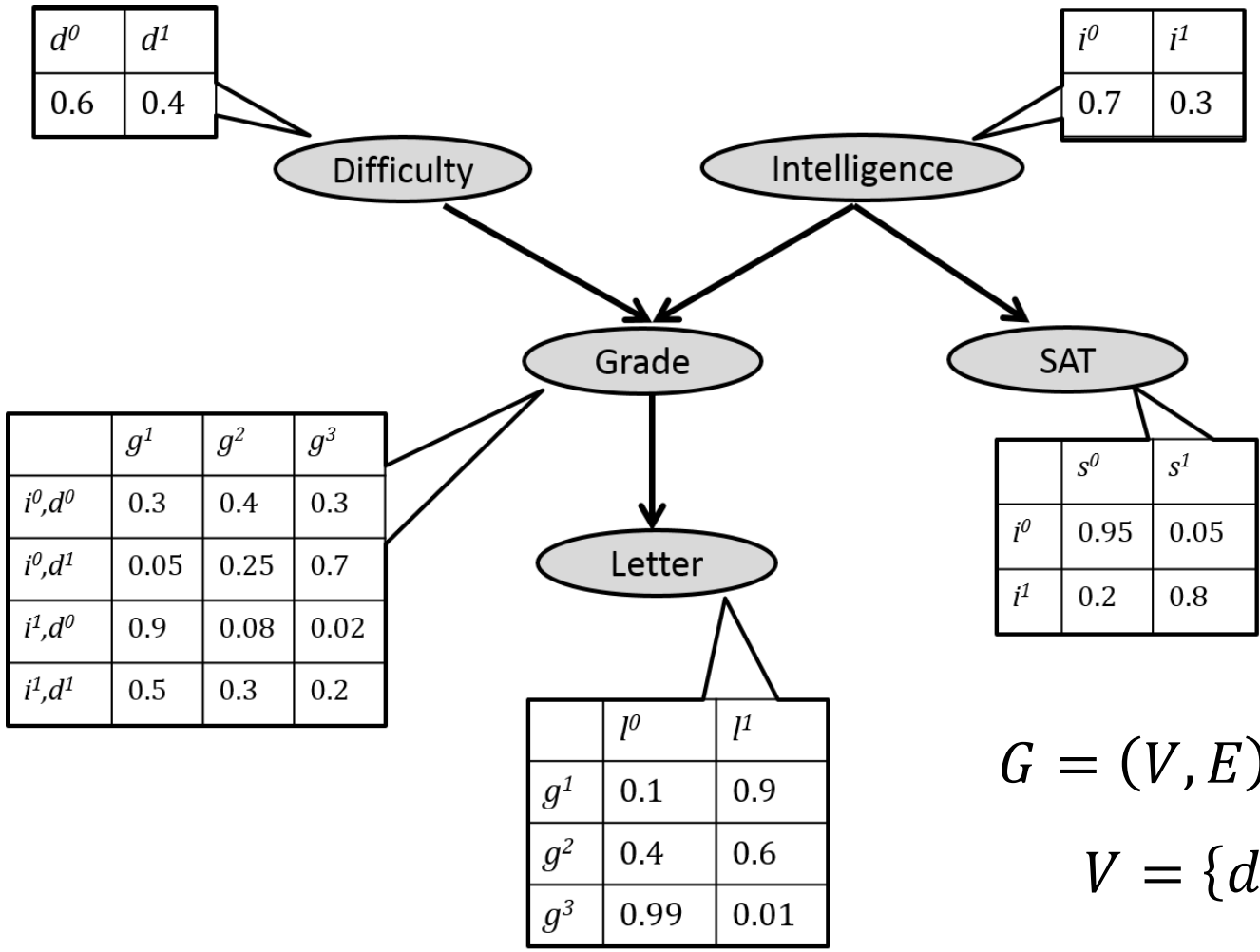
DAG of a first-order Markov Chain

Probabilistic Models of the Data

- **Bayesian Networks: A Toy Example**

Prior Distribution

$p(x_i)$



d^0	d^1
0.6	0.4

i^0	i^1
0.7	0.3

	g^1	g^2	g^3
i^0,d^0	0.3	0.4	0.3
i^0,d^1	0.05	0.25	0.7
i^1,d^0	0.9	0.08	0.02
i^1,d^1	0.5	0.3	0.2

	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

	l^0	l^1
g^1	0.1	0.9
g^2	0.4	0.6
g^3	0.99	0.01

CPT: Conditional Probability Table

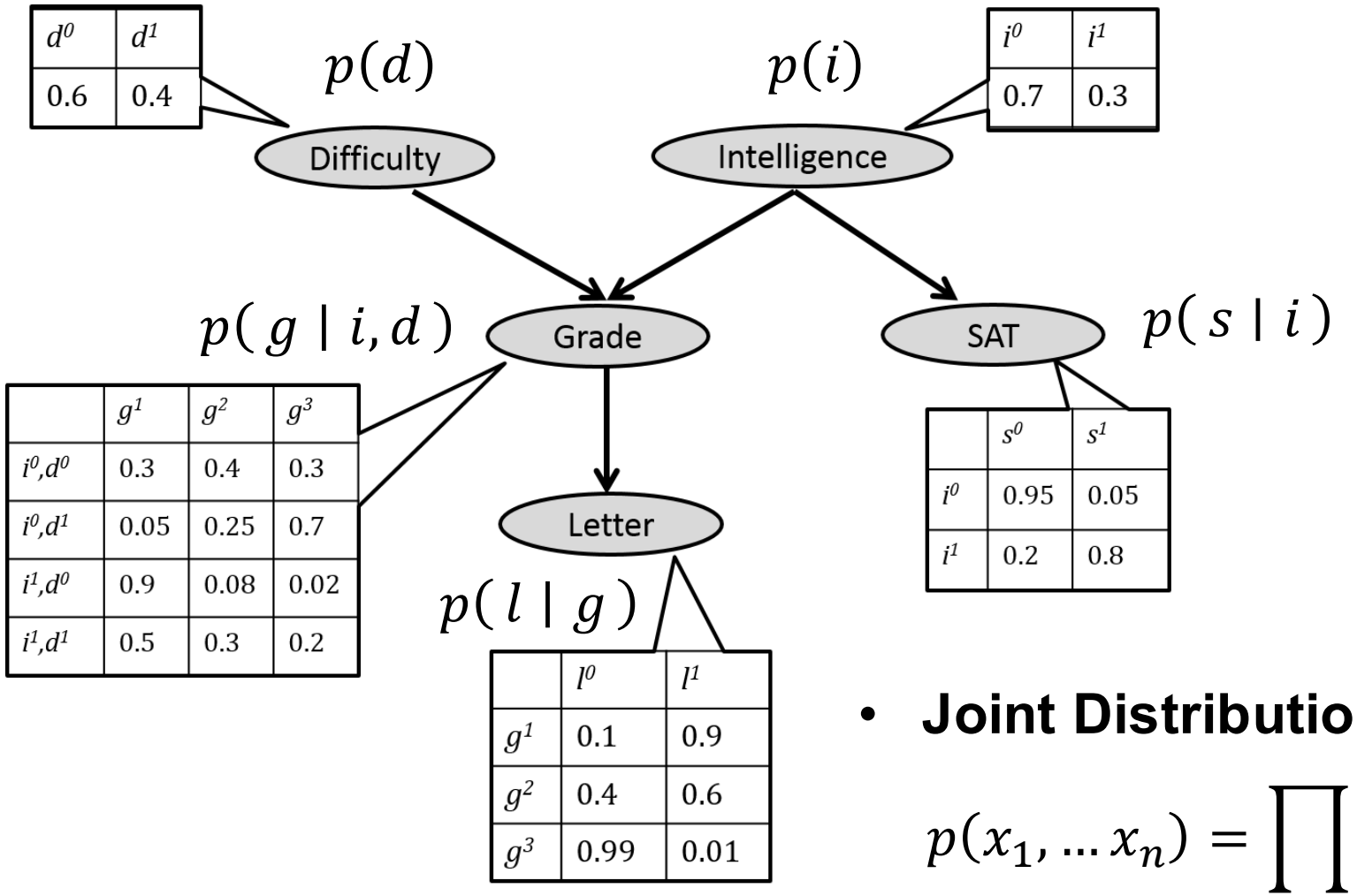
Conditional Distribution

$p(x_i \mid \boldsymbol{x}_{Pa(i)})$

$$G = (V, E), |V| = 5$$

$$V = \{d, i, g, s, l\}$$

Probabilistic Models of the Data



d^0	d^1
0.6	0.4

i^0	i^1
0.7	0.3

	g^1	g^2	g^3
i^0,d^0	0.3	0.4	0.3
i^0,d^1	0.05	0.25	0.7
i^1,d^0	0.9	0.08	0.02
i^1,d^1	0.5	0.3	0.2

	s^0	s^1
i^0	0.95	0.05
i^1	0.2	0.8

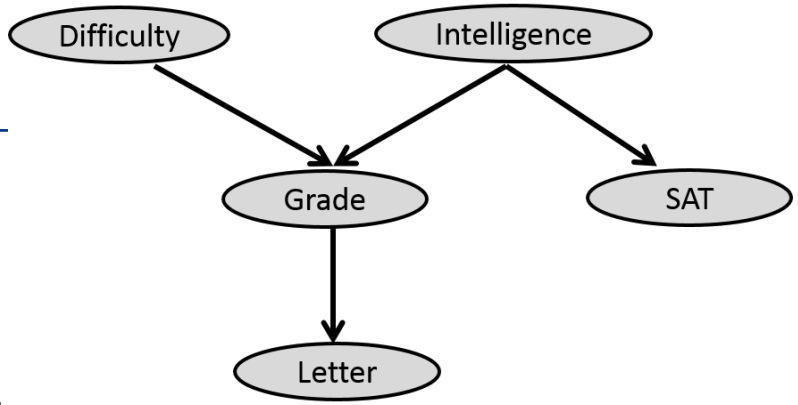
	l^0	l^1
g^1	0.1	0.9
g^2	0.4	0.6
g^3	0.99	0.01

- **Joint Distribution Formulation:**

$$p(x_1, \dots x_n) = \prod_{i \in V} p\left(x_i \mid \boldsymbol{x}_{Pa(i)}\right)$$

$$p(d,i,g,s,l) = p(d)\, p(i)\, p(g \mid i,d)\, p(s \mid i)\, p(l \mid g)$$

Probabilistic Models of the Data



- **Bayesian Networks: A Toy Example**
  - Factorization corresponding to the **assumed** DAG

$$p(d,i,g,s,l) = p(d)\,\boldsymbol{p(i)}\,p(g \mid i,d)\,\boldsymbol{p(s \mid i)}\,\boldsymbol{p(l \mid g)}$$

  - A legal factorization obtained by chain rule

$$p(d,i,g,s,l) = p(d)\,\boldsymbol{p(i \mid d)}\,p(g \mid i,d)\,\boldsymbol{p(s \mid i,d,g)}\,\boldsymbol{p(l \mid g,d,i,s)}$$

  - Thus, the following **extra independencies** are imposed by the DAG

$$\boldsymbol{D \perp I} \qquad \boldsymbol{S \perp \{D,G\} \mid I} \qquad \boldsymbol{L \perp \{I,D,S\} \mid G}$$

Summary: Probabilistic Models of the Data

- **Representation of $p(x)$**
  - **Joint Distribution:** Direct modeling $p(x_1, \dots, x_n)$ is **unaffordable**

    ➔ **Exponential** parameters $(k^n - 1)$ ➔ "Curse of Dimensionality"

  - **Conditional Independence Assumptions:** Truncate the conditioning set, thus exponentially reduce parameters

$$p(x_1, \dots, x_n) = \prod_i p(x_i \mid x_{A_i}) \qquad A_i \subseteq \{x_1, \dots, x_{i-1}\}$$

  - **Bayesian networks:** represent the factorization with graph $BN = (G, P)$
    - ✓ $A_i = \text{Parents}(x_i)$ in the DAG
    - ✓ **Graph structure encodes independence assumptions**

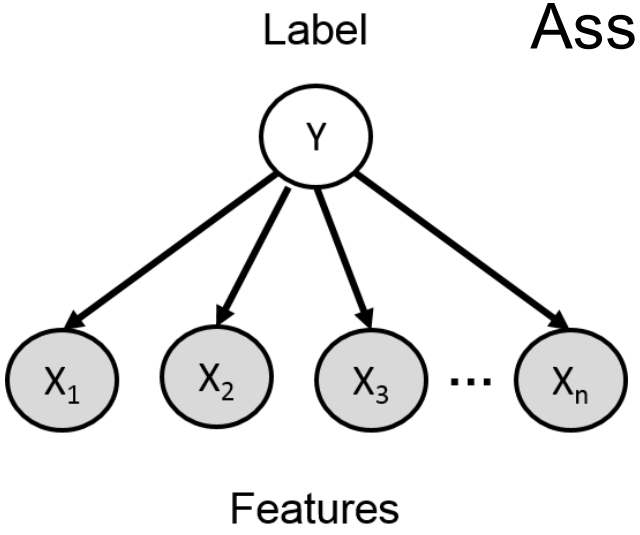


Lecture Outline

- **Probabilistic Models of the Data**
- **Generative vs. Discriminative Models**
- **Deep Generative Models**

Naïve Bayes: Generative Classification

- **E-mail: Spam ($Y = 1$) or not Spam ($Y = 0$)**
  - Let $1{:}\,n$ index the words in the vocabulary
  - $X_i = 1$ if the ith word appears in an e-mail ($X_i = 0$ otherwise)
  - E-mails follow a joint distribution $p(Y, X_1, \ldots, X_n)$
- **Naïve Bayes**



Assume words are conditionally independent given Y

$$p(y, x_1, \ldots x_n) = p(y) \prod_{i=1}^{n} p(x_i \mid y)$$

Is $p(y, x_1, \ldots x_n)$ what we truly want?

Naïve Bayes: Generative Classification

- **Spam Classification**
  - **Goal:** Learn a classification model $p(y \mid x_1, \dots x_n)$
  - **Bayes' Rule:**

$$p(Y = 1 \mid x_1, \dots x_n) = \frac{p(Y = 1, x_1, \dots x_n)}{p(x_1, \dots x_n)} = \frac{p(x_1, \dots x_n \mid Y = 1)\, p(Y = 1)}{p(x_1, \dots x_n)}$$

  - ✓ $p(Y = 1)$: Prior distribution ➔ can be set in advance
  - ✓ $p(x_1, \dots x_n)$: Data distribution ➔ cancels when comparing classes
  - ✓ $p(x_1, \dots x_n \mid Y = 1)$: Conditional data distribution ➔ Generative Modeling

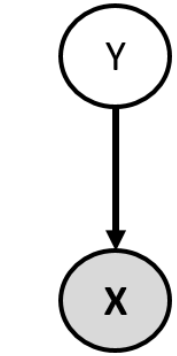
If conditionally independent:

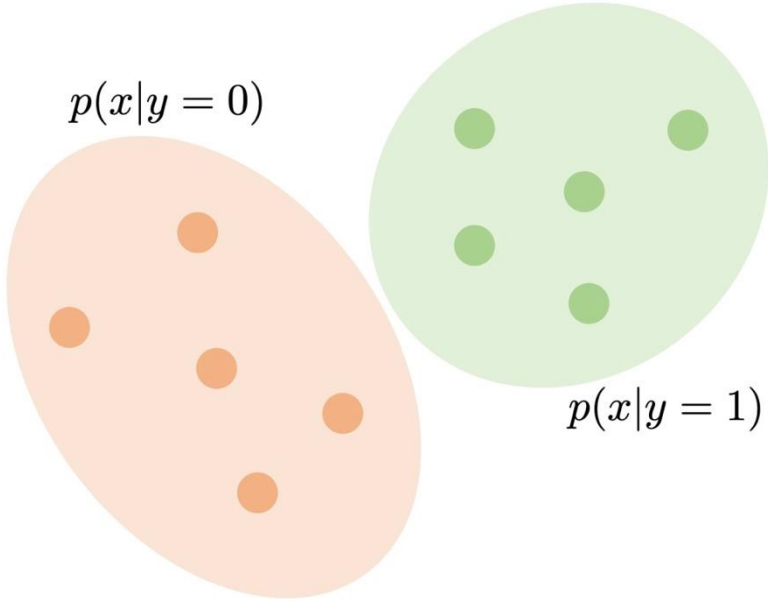
$$p(x_1, \dots x_n \mid Y = 1) = \prod_{i=1}^{n} p(x_i \mid Y = 1)$$

Generative Models vs. Discriminative Models

We want to predict $p(Y \mid \boldsymbol{X})$

Generative

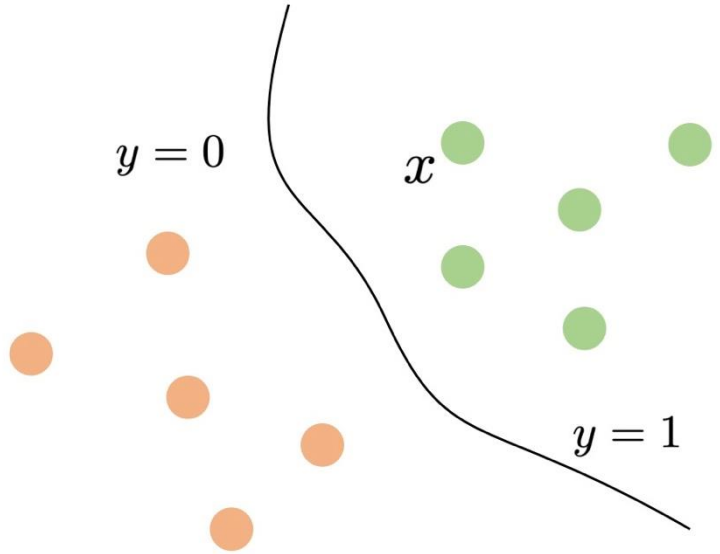




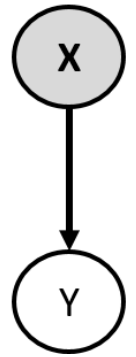
Learn $p(Y)$ and $p(\boldsymbol{X} \mid Y)$

Bayes' Rule

$p(Y \mid \boldsymbol{X})$



Discriminative



Directly learn $p(Y \mid \boldsymbol{X})$

- No need to learn distribution $p(\boldsymbol{X})$
- Only works in discriminative tasks

Why?

Generative Models vs. Discriminative Models

- Generative models apply to discriminative tasks: **Bayes' rule**

$$p(y|x) = p(x|y)\frac{p(y)}{p(x)}$$

discriminative — generative

assuming known prior

constant for given x

- Can discriminative models be generative?

$$p(x|y) = p(y|x)\frac{p(x)}{p(y)}$$

generative — discriminative

still need to model prior distribution of x

constant for given y

- The key challenge lies in modeling the data distribution

Discriminative Models

- **Example: Logistic Regression**
  - **Formulation:** use a function $f(\boldsymbol{x}, \boldsymbol{\alpha})$ to approach $p(Y = 1)$

$$p(Y = 1 \mid \boldsymbol{x}; \boldsymbol{\alpha}) = f(\boldsymbol{x}, \boldsymbol{\alpha})$$

  ➔ Modeling $p(Y \mid \boldsymbol{X})$ through a parameterized representation

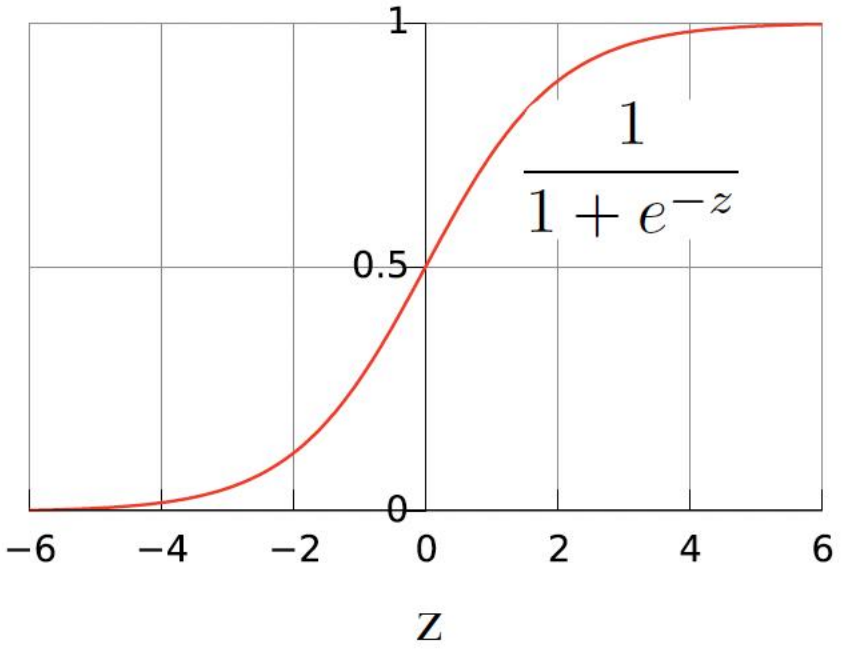
  - **Linear Dependence:**

$$z(\boldsymbol{x}, \boldsymbol{\alpha}) = \alpha_0 + \sum \alpha_i x_i$$

No Independence Assumption

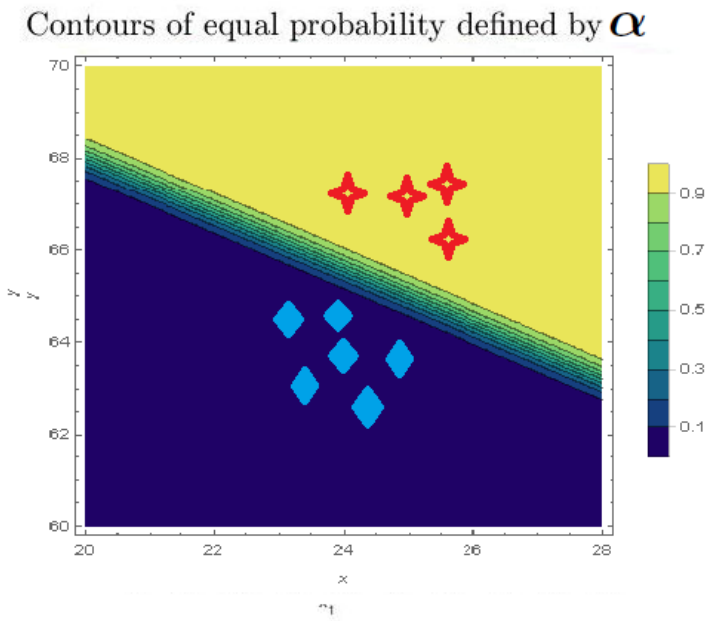
$$p(Y = 1 \mid \boldsymbol{x}; \boldsymbol{\alpha}) = \frac{1}{1 + e^{-z(\boldsymbol{x}, \boldsymbol{\alpha})}}$$

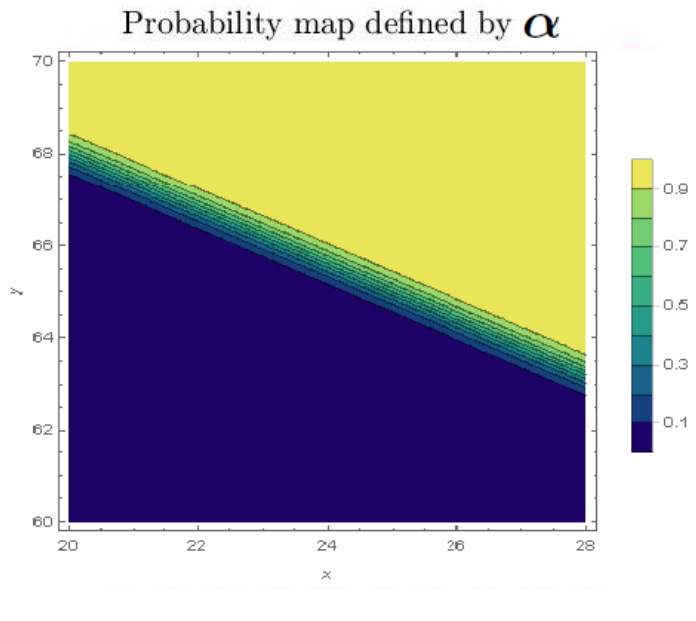
Logistic Function

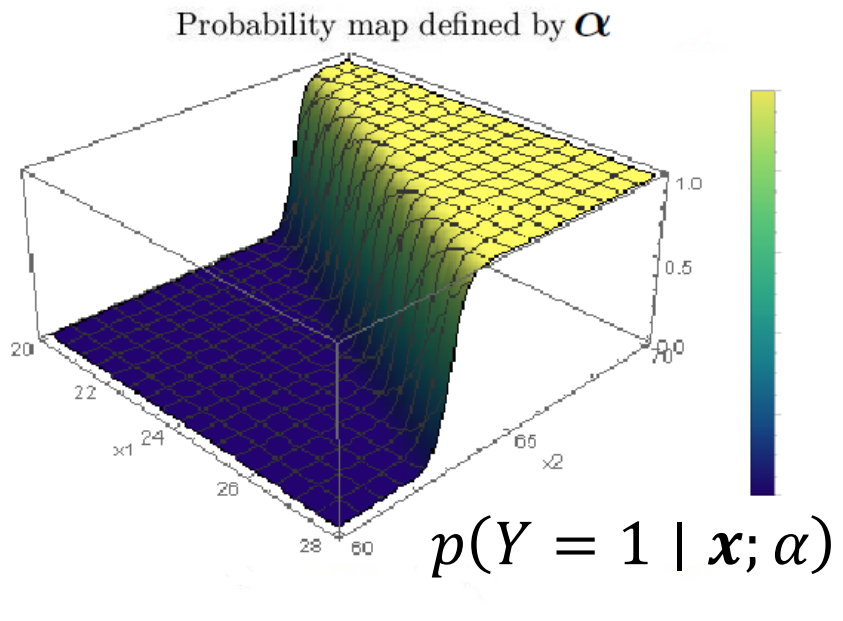


Discriminative Models

- **Example: Logistic Regression**
  - ➢ Linear dependence determine the property of decision boundary







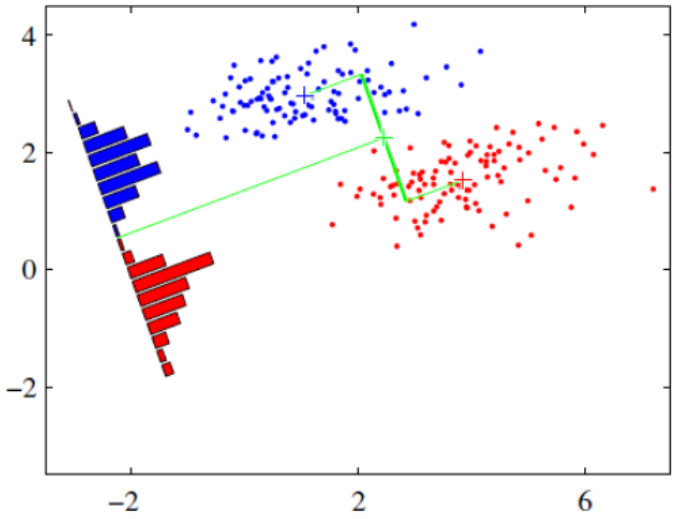
$p(Y = 1 \mid \boldsymbol{x}; \alpha)$

  - ➢ The Equal Probability Contour ➔ 'Decision boundary'
  - ➢ Decision boundary $p(Y = 1 \mid \boldsymbol{x}; \boldsymbol{\alpha}) = p(Y = 0 \mid \boldsymbol{x}; \boldsymbol{\alpha})$ is linear

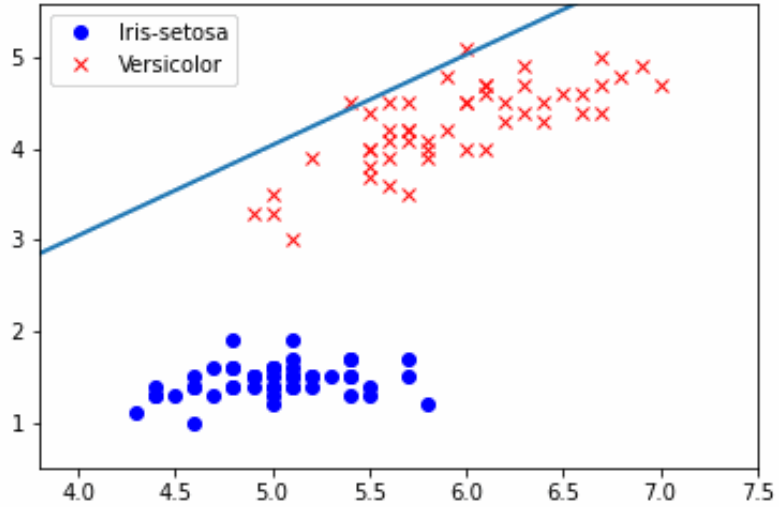
Discriminative Models

- **Linear discriminative models**
  - ➢ Logistic Regression use a function $f(\boldsymbol{x}, \boldsymbol{\alpha})$ to approach $p(Y = 1 \mid \boldsymbol{x}; \alpha)$
    - ➔ **Not necessary, could directly formulate the decision boundary**

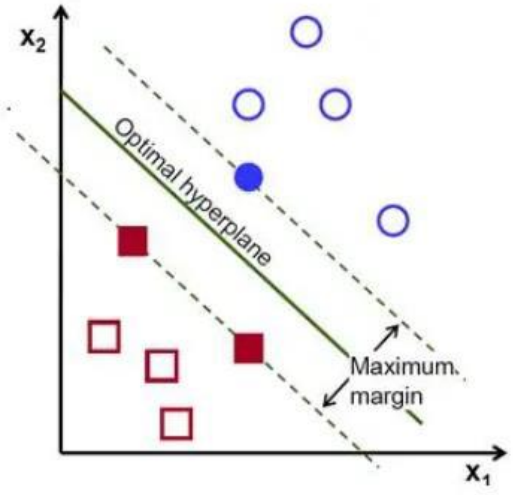
$$f(\boldsymbol{x}) = \boldsymbol{w}^{\top}\boldsymbol{x} + b$$



Linear Discriminant Analysis



Perceptron



Support Vector Machine



Lecture Outline

- **Probabilistic Models of the Data**
- **Generative vs. Discriminative Models**
- **Deep Generative Models**

Neural Networks

- **Data Dependence Modeling**
  - **Linear Dependence:** In logistic regression, we assume

$$p(Y = 1 \mid \boldsymbol{x}; \boldsymbol{\alpha}) = f(\boldsymbol{x}, \boldsymbol{\alpha})$$

where $z(\boldsymbol{x}, \boldsymbol{\alpha}) = \alpha_0 + \sum \alpha_i x_i$ and $p(Y = 1 \mid \boldsymbol{x}; \boldsymbol{\alpha}) = \sigma\big(z(\boldsymbol{x}, \boldsymbol{\alpha})\big)$

➔ **Linear dependence modeling might be too simple**

  - **Non-Linear Dependence:**
    - ✓ Let $h(\boldsymbol{A}, \boldsymbol{b}, \boldsymbol{x}) = g(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b})$ be a non-linear transformation of input $\boldsymbol{x}$

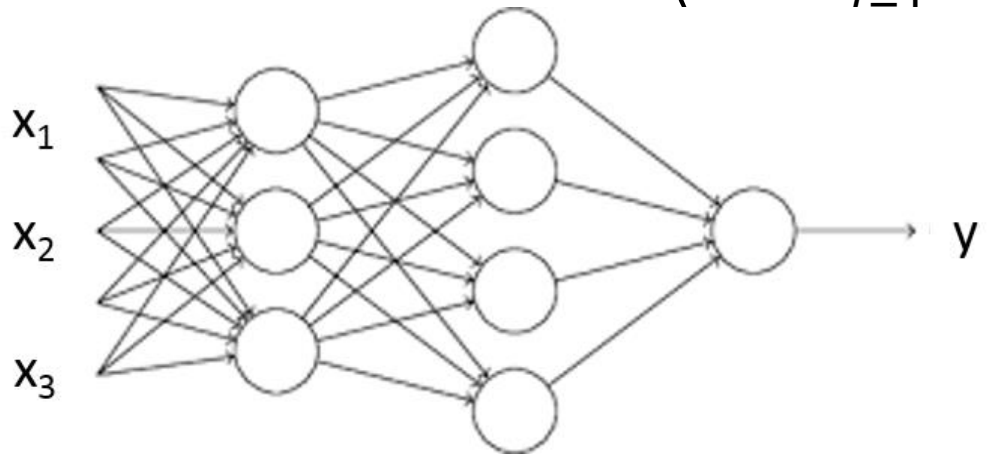
$$p_{Neural}(Y = 1 \mid \boldsymbol{x}; \boldsymbol{\alpha}, \boldsymbol{A}, \boldsymbol{b}) = \sigma\left(\alpha_0 + \sum_{i=1}^{h} \alpha_i h_i\right)$$

More flexible with more params.

Neural Networks

- **Data Dependence Modeling**
  - ➢ **Non-Linear Dependence:**
    - ✓ Let $h(\boldsymbol{A}, \boldsymbol{b}, \boldsymbol{x}) = g(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b})$ be a non-linear transformation of input $\boldsymbol{x}$

$$p_{Neural}(Y = 1 \mid \boldsymbol{x}; \boldsymbol{\alpha}, \boldsymbol{A}, \boldsymbol{b}) = \sigma\left(\alpha_0 + \sum_{i=1}^{h} \alpha_i h_i\right)$$



Can stack to multiple layers for higher non-linearity

Bayesian Networks vs. Neural Networks

- **Chain Rule**

$$p(x_1, x_2, x_3, x_4) = p(x_1)\, p(\, x_2 \mid x_1\,)\, p(\, x_3 \mid x_1, x_2\,)\, p(\, x_4 \mid x_1, x_2, x_3\,)$$

  - **Pros:** no assumption required; **Cons:** exponential amount of parameters

- **Bayes Networks** *CPT: Conditional Probability Table*

$$p(x_1, x_2, x_3, x_4) \approx p_{CPT}(x_1)\, p_{CPT}(\, x_2 \mid x_1\,)\, p_{CPT}(\, x_3 \mid \cancel{x_1}, x_2\,)\, p_{CPT}(\, x_4 \mid x_1, \cancel{x_2}, \cancel{x_3}\,)$$

  - Assumes conditional independencies ➔ simplify CPT

- **Neural Networks**

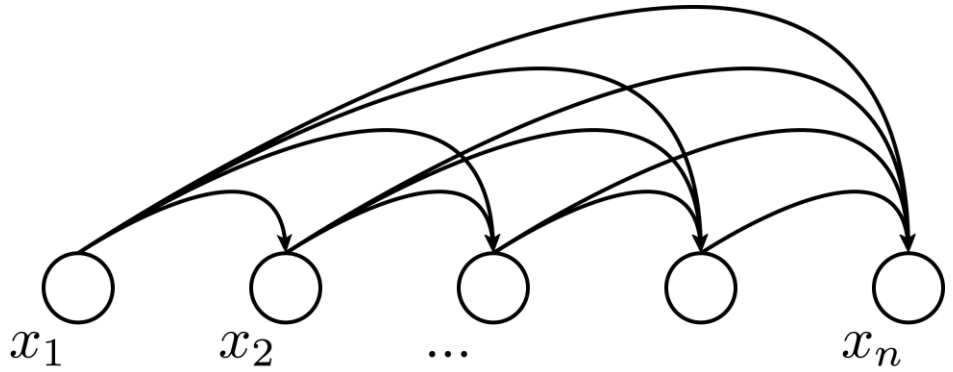
$$p(x_1, x_2, x_3, x_4) \approx p(x_1)\, p(\, x_2 \mid x_1\,)\, p_{Neural}(\, x_3 \mid x_1, x_2\,)\, p_{Neural}(\, x_4 \mid x_1, x_2, x_3\,)$$

  - **Parametrize** the conditionals, instead of simplify them ➔ $O(n^2)$ if linear

Bayesian Networks vs. Neural Networks

- **Neural Networks: Case Study**
  - ➢ Factorization defined on **input representation space** X
    - ✓ Autoregressive models

$$p(x_1, x_2, ..., x_n) = p(x_1)p(x_2 \mid x_1)...p(x_n \mid x_1, x_2, ..., x_{n-1})$$

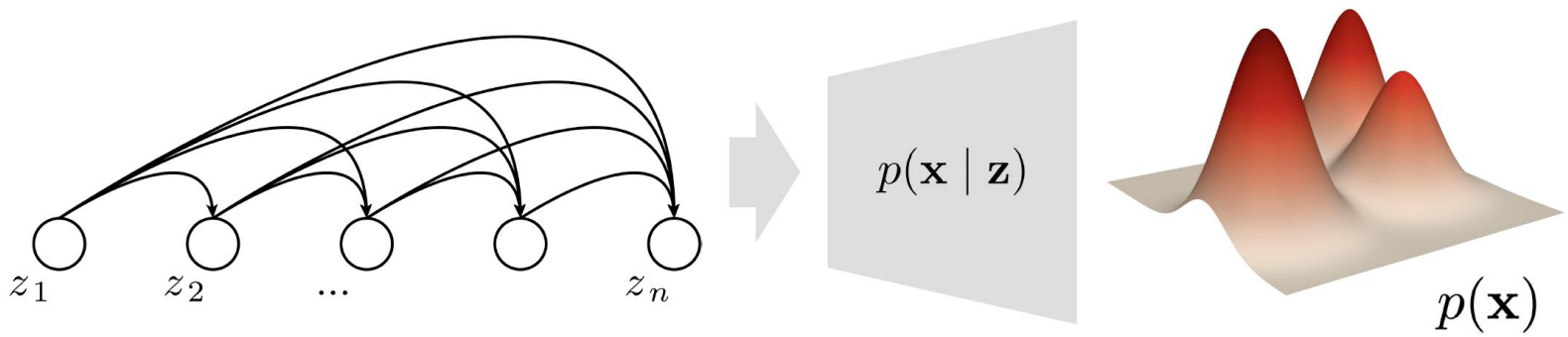


x_i can be text tokens or pixels

Bayesian Networks vs. Neural Networks

- **Neural Networks: Case Study**
  - ➢ Factorization defined on **latent representation space** Z
    - ✓ Autoregressive on VQ-VAE tokens

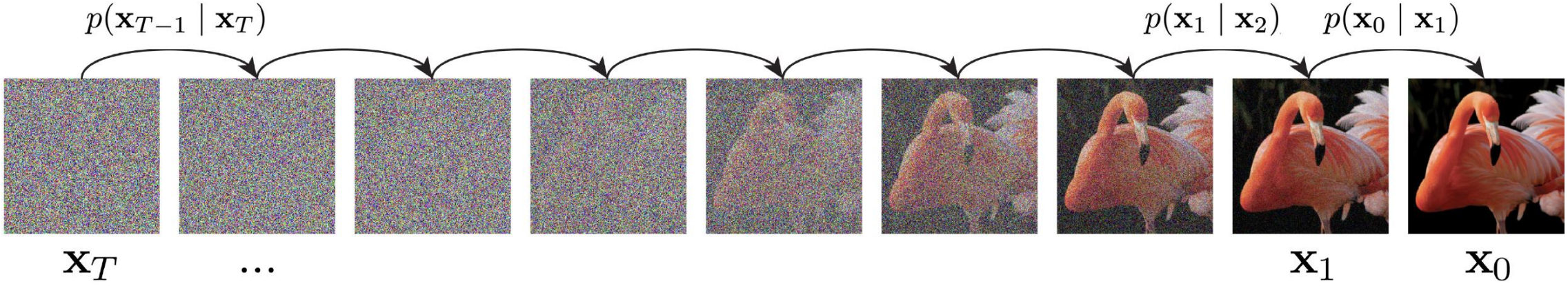
$$p(\mathbf{x}, \mathbf{z}) = p(\mathbf{z})p(\mathbf{x} \mid \mathbf{z})$$
$$\text{with} \quad p(\mathbf{z}) = p(z_1)p(z_2 \mid z_1)...p(z_n \mid z_1, z_2, ..., z_{n-1})$$



Bayesian Networks vs. Neural Networks

- **Neural Networks: Case Study**
  - ➢ Factorization defined on **distribution transformation process**
    - ✓ Diffusion Models

$$p(\mathbf{x}_{0:T}) = p(\mathbf{x}_T)p(\mathbf{x}_{T-1} \mid \mathbf{x}_T)...p(\mathbf{x}_1 \mid \mathbf{x}_2)p(\mathbf{x}_0 \mid \mathbf{x}_1)$$



Factorization vs. Parameterization

- **Same factorization $Z \to X$, $p_{Z,X}(z,x) = p_Z(z)\, p_{X|Z}(x \mid z)$**
  - **Mixture of 2 Gaussians**
    - ✓ $Z \sim \text{Bernoulli}(p)$
    - ✓ $X \mid (Z=0) \sim \mathcal{N}(\mu_0, \sigma_0)$, $X \mid (Z=1) \sim \mathcal{N}(\mu_1, \sigma_1)$

    Parameters: $p, \mu_0, \sigma_0, \mu_1, \sigma_1$
  - **Mixture of Infinite Gaussians**
    - ✓ $Z \sim \mathcal{U}(a,b)$, $X \mid (Z=z) \sim \mathcal{N}(z, \sigma)$

    Parameters: a, b, σ
  - **Mixture of Infinite Gaussians**
    - ✓ $Z \sim \mathcal{N}(0,1)$
    - ✓ $X \mid (Z=z) \sim \mathcal{N}\left(\mu_\theta(z), e^{\sigma_\phi(z)}\right)$, where μ_θ and σ_ϕ are neural networks

    Parameters: θ, ϕ



Summary: Lecture for Today

- **Probabilistic Models of the Data**
  - Examples of Probabilistic Models
  - The Curse of Dimensionality
  - Parameter-Efficient Models Through Conditional Independence
  - Bayesian Networks: An Example of Shallow Generative Models
- **Generative vs. Discriminative Models**
  - Naive Bayes vs. Logistic Regression
- **Deep Generative Models**

Supplementary Material

- **Stanford CS 236 Lecture 2**
  - https://www.youtube.com/watch?v=rNEujZmD2Tg&list=PLoROMvodv4rPOWA-omMM6STXaWW4FvJT8&index=2
- **Cornell CS 6785 Lecture 2**
  - https://www.youtube.com/watch?v=FBGj_B6hH9Y&list=PL2UML_KCiC0UPzjW9BjO-IW6dqliu9O4B&index=2
- **Deep Generative Models**
  - **Section 2**

Reading List

- **Reading List:**
  - D. Zoran and Y. Weiss, From Learning Models of Natural Image Patches to Whole Image Restoration, ICCV 2011
  - D. Zoran and Y. Weiss, Natural Images, Gaussian Mixtures and Dead Leaves, NeurIPS 2012
  - D. Ulyanov et al., Deep Image Prior, CVPR 2018
- **(Optional) Recommended readings:**
  - A. A. Efros and T. K. Leung, Texture Synthesis by Non-parametric Sampling, 1999
  - C. Barnes et al., PatchMatch: A Randomized Correspondence Algorithm for Structural Image Editing, 2009
  - W. T. Freeman et al., Example-Based Super-Resolution, 2002